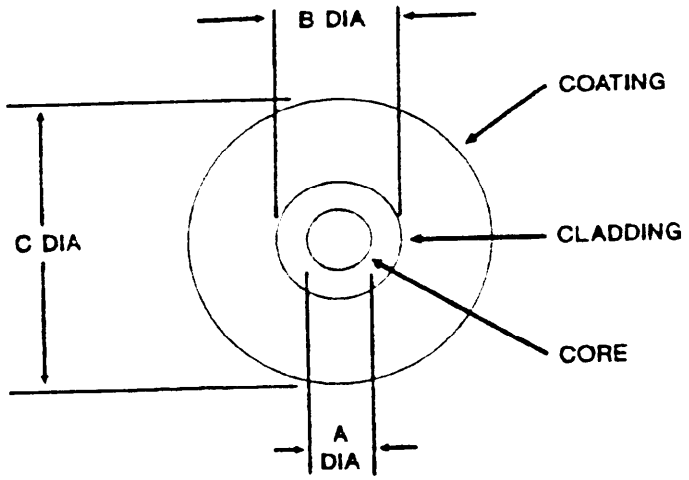
| METRIC |
MIL-F-49291/1A
8 July 1992
SUPERSEDING
DOD-F-49291
30 MARCH 1987

MILITARY SPECIFICATION SHEET

FIBER, OPTICAL, 50/125 MICROMETERS, RADIATION HARDENED (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the optical fibers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-49291

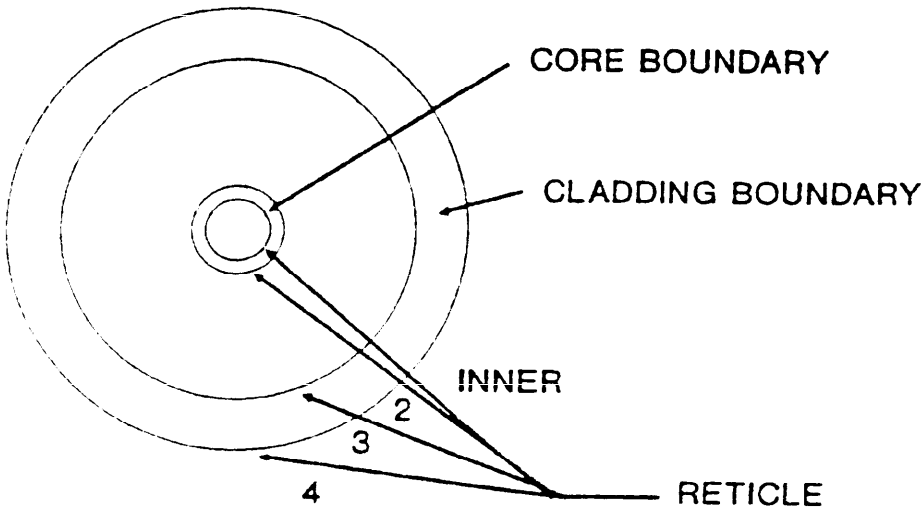


Fiber index	PIN	Dimensions		
		A	B	C
Graded	M49291/1-01	50 ± 3	125 ± 2	250 ± 15

NOTE: Dimensions are in micrometers.

FIGURE 1. Dimensions and configuration of optical fiber construction.

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Circle	Diameter of circle
Inner	47 Micrometers
Second	53 Micrometers
Third	123 Micrometers
Fourth	127 Micrometers

NOTE: Dimensions are in micrometers.

FIGURE 2. Tolerance fields.

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REQUIREMENTS:

Classification:
Type: I.
Class: 1.
Composition: A.
Size: III.
Wavelength: C.

Dimensions and configuration:
Diameter: See Figures 1 and 2.
Ovality:
Core: 10 percent
Cladding: 4 percent
Offset:
Core-to-cladding: $\leq$ 3 micrometers.
Fiber-to-coating: $\leq$ 17.5 micrometers (OCCR $\geq$ 0.56)
Maximum percent of coating diameter change at the splice points: Not applicable.
Continuous lengths: 1100 meters minimum.

Splices: Not allowed.

Tensile strength (proof test): 690 MPa.

Fiber mass/unit length (kg/km): 0.1 kg/km maximum.

Attenuation rate: Measurements made at 1300 nm $\pm$ 20 nm.

	-46°C	+25°C	+71°C
Non-nuclear:	1.25dB/km	1.25dB/km	1.25dB/km
After each required nuclear radiation resistance test:	3.25dB/km	2.75dB/km	2.25dB/km

Numerical aperture: 0.23 $\pm$ 0.02 at 850 nanometers $\pm$ 20 nm.

Bandwidth: 400 MHz·km at 1300 nm $\pm$ 20 nm

Fluid immersion aging: Not applicable.

Storage temperature: Applicable.

Temperature range:
Operating: -46°C to +71°C.
Storage: -57°C to +85°C.

Fungus test: Applicable.

Life aging: Applicable, except that conditions shall be + 110°C for 240 hours.

Nuclear radiation resistance: The nuclear radiation resistance characteristics of this optical fiber are classified and are contained in an appendix to this specification. Application to receive this appendix must be made through the US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5016. Information concerning security clearance classification and "need to know" must be detailed in the request.

Mechanical strippability: Applicable, except the requirements of EIA/TIA-455-178 shall not apply.

Attenuation uniformity: Not Applicable.

Macrobend attenuation: Not Applicable.

Dispersion: Not Applicable.

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PIN (see Figure 1 and table I): M49291/01-01

TABLE I. Supersession data.

PIN	Superseding
M49291/1-01	D49291/1-01
M49291/1-02	None

Qualification inspection shall consist of performing the inspections and optical tests specified in Table I herein, in the sequence shown.

Group A inspection shall consist of performing the inspections and optical tests specified in Table II herein, in the sequence shown.

Group B inspection shall consist of performing the inspections and optical tests specified in Table III herein, in the sequence shown.

Group C inspection shall consist of performing the inspections and optical tests specified in Table IV herein, in the sequence shown.

TABLE 1. Qualification inspection.

Inspection	Specimen length	Sample Quantity
Group I		
Visual and mechanical	1 km	10 units
Attenuation rate	1 km	10 units
Numerical aperture	2 m	10 units
Core Diameter	2 m	10 units
Bandwidth by time domain	1 km	10 units
Group II		
Proof test	1 km	5 units
Mechanical strippability	2 m	5 units
Group III		
Temperature cycling	1 km	5 units
Temperature shock	1 km	5 units
Temperature-humidity cycling	1 km	3 units
Nuclear radiation resistance	As necessary	36 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ Specified units are derived from 4 each, 1 km optical fibers. Units from each fiber are to be subjected to each of the three required tests (prompt gamma, neutron and residual gamma) at each test temperature (-46°C, +25°C, and 71°C). The testing of multiple sample units at a time is permissible.

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TABLE II. Group A inspection.

Inspection	Specimen length
Visual and mechanical	1 km
Attenuation rate	1 km
Numerical aperture	2 m
Core Diameter	2 m
Bandwidth by time domain	1 km

TABLE III. Group B inspection.

Inspection	Specimen length
Proof test	1 km
Mechanical strippability	2 m

TABLE IV. Group C inspection.

Inspection	Specimen length	Sample Quantity
Subgroup I (3 month interval)		
Temperature cycling	1 km	4 units
Subgroup II (6 month interval)		
Temperature shock	1 km	5 units
Temperature-humidity cycling	1 km	3 units
Subgroup III (12 month interval)		
Nuclear radiation resistance	As necessary	12 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ Specified units are derived from 4 each, 1 km optical fibers. Units from each fiber are to be subjected to only the residual gamma at a test temperature of -46°C. The testing of multiple sample units at a time is permissible.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI, SC
Navy - AS
DLA - ES

User activity:
Air Force

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6010-0036-1)

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